Semester:	Year I Semester I
Course Code:	PMT 1013
Course Name:	Foundations of Mathematics
Credit Value:	3.0
Core/Optional:	Core

Introduction

This course is designed to develop your ability to think logically, construct rigorous arguments, and understand the fundamental structures that support modern mathematics.

We will begin with logical reasoning, learning how to analyze statements, use formal logic, and understand the role of quantifiers and logical equivalence. These tools will lead us into the art of mathematical proofs, where you will learn various techniques such as direct proof, contradiction, induction, and more.

As the course progresses, we will explore key mathematical structures including sets, relations, and functions, which form the language of mathematics. Finally, we will study axiomatic systems, focusing on the natural numbers and real numbers, and the principles that govern them.

By the end of this course, you will not only understand these core concepts but also be able to apply rigorous reasoning to solve problems and prove mathematical results with confidence.

Assessment Strategy:	
Continuous Assessments (Mid-Exams) 30%	Final Assessment (End-Exam) 70%

Recommended reading:

- Velleman, Daniel J. How to Prove It: A Structured Approach, 2nd Edition, Cambridge University Press, 2006.
- Hammack, Richard. Book of Proof. 3rd Edition, CreateSpace Independent Publishing Platform, 2013.
- Carroll, John B. Introduction to Abstract Mathematics, 2012
- Cummings, Jay. Proofs: A Long-form Mathematics Textbook. Independently published, 2021.

Introduction to Modern Mathematics

The field of mathematics emerged from the human need to count items, determine areas, and understand the natural world. The word itself comes from the Greek words *mathema* ("science, knowledge, or learning") and *mathematikos* ("fond of learning").

Ancient Greek mathematicians are credited with laying the foundation for modern mathematics because they:

- Were the first to think **abstractly** about the subject.
- Developed the practice of recording logical arguments in **proofs**.
- Studied **pure mathematics**, whereas earlier civilizations like the Babylonians and Egyptians focused on practical applications like trade.

To explain something **"abstractly about the subject"** means to talk about the **general idea or concept** without focusing on specific examples, numbers, or detailed situations.

It is like discussing the **main principle** rather than a particular case.

Simple Meaning

- **Concrete explanation** → specific and detailed
- **Abstract explanation** → general and conceptual

Example 1: Triangle

Concrete:

"The triangle ABC has sides 3 cm, 4 cm, and 5 cm."

Abstract:

"A triangle is a polygon with three sides, and the sum of its interior angles is always 180°."

Here, we discuss the idea of triangles in general, not one particular triangle.

Example 2: Even Numbers

Concrete:

"8 is an even number because it is divisible by 2."

Abstract:

"An even number is any integer of the form $2n$, where n is an integer."

This abstract form gives the rule for all even numbers. $n = 2k$

The Structure of Modern Mathematics

"Modern mathematics" refers to the current formal axiomatic system based on rigorous logical foundations. Unlike its roots in simple counting and geometry, modern mathematics or **axiomatic mathematics** is the science of operations on collections of arbitrary objects.

It follows a specific developmental structure:

Axioms $\Longrightarrow$ Definitions $\Longrightarrow$ Conjectures $\Longrightarrow$ Proofs $\Longrightarrow$ Theorems $\Longrightarrow$ Generalizations and Extensions

Conjecture:
A statement in mathematics that is believed to be true based on examples or patterns, but has not yet been proved.

1. Goldbach's Conjecture

Statement:
Every even number greater than 2 can be written as the sum of two prime numbers.

Examples:
4 = 2 + 2
8 = 3 + 5
10 = 5 + 5
28 = 11 + 17

It has been tested for very large numbers, but no complete proof exists yet.

2. Twin Prime Conjecture

Statement:
There are infinitely many pairs of prime numbers that differ by 2.

Examples:
(3, 5)
(5, 7)
(11, 13)
(17, 19)

These are called twin primes.

3. Collatz Conjecture

Statement:
Take any positive integer:

if it is even, divide by 2
if it is odd, multiply by 3 and add 1

Repeat the process. The conjecture says you will always eventually reach 1.

Example:

$6 \to 3 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1$

Still unproved.

Inductive and Deductive Reasoning

Early mathematical discoveries were often motivated by physical scientists observing the physical world. This method of making inferences based on observations is known as **inductive reasoning**.

- **Definition:** Inductive reasoning is the method of reasoning based on making inferences and conclusions from observations.

Key Concepts

- **Definition:** Inductive reasoning uses a particular set of observations to extrapolate a broader conclusion.
- **Example:** Predicting the sun will rise tomorrow simply because it has risen every day in the past.
- **Historical Context:** Early mathematics relied heavily on induction and observed results, largely because it was driven by the study of physical phenomena.

Inductive Reasoning in Mathematics

Induction is useful for developing **conjectures**, it cannot serve as absolute proof.

- **The Fallibility of Induction:** A classic example is Pierre de Fermat's conjecture that $2^{2^n} + 1$ is prime for all natural numbers n.
- **The Counterexample:** While true for $n = 1, 2, 3, 4$, Leonhard Euler later proved it false by showing that $2^{2^5} + 1$ is not prime.

  4,294,967,297=641×6,700,417

- The human brain is great at spotting patterns, but patterns can be misleading. For example, the formula $n^2 - n + 41$ produces prime numbers for every from 1 to 40, but it fails at $n = 41$.
  Principle of Mathematical Induction provides a **rigorous mathematical guarantee** that a pattern will never break, no matter how high you count.

- **Conclusion:** Data-based reasoning can support a conjecture, but only formal proof provides absolute certainty in mathematics.

Definition and History

Deductive reasoning is defined as a method where a conclusion is reached through logical steps based on established assumptions.

Historically, this approach was pioneered by Greek mathematicians:

- **Thales, Pythagoras, and Aristotle** used it to justify mathematical results.
- **Euclid** is famously credited with using deduction to prove there are infinitely many prime numbers.
- A student of Pythagoras used it to prove the **irrationality of** $\sqrt{2}$.

Example 1: Let $x = 0.\bar{9}$. Then, using deductive reasoning, it can be shown that the conjecture $x = 1$ is true. In fact, there are many different ways to show deductively that $x = 0.\bar{9} = 1$, including the following deductive argument.

Let $x = 0.\bar{9}$. Then $10x = 9.\bar{9}$

$$10x - x = 9x = 9$$
$$9x = 9$$
$$x = 1$$

A classic example of deductive reasoning is the proof that the **sum of two even integers is always even**. This uses a logical progression based on the formal definitions of numbers.

The Proof

1. Definitions (Assumptions):

- An integer n is **even** if it can be written as $2k$, where k is any integer.
- Let our two even integers be a and b.

2. Logical Steps:

- By definition, there exist integers m and n such that $a = 2m$ and $n = 2n$.
- Now, we look at their sum:
  $a + b = 2m + 2n$.
- Using the distributive property (factoring out the 2):
  $a + b = 2(m + n)$

3. Conclusion:

- Since m and n are integers, their sum $(m + n)$ must also be an integer. Let's call this new integer p.
- Therefore, $a + b = 2p$.
- Because the sum fits the definition of an even number $(2 \times integer)$, the sum $a + b$ is **even**.

To prove that the interior angles of a triangle sum to 180^0, we use deductive reasoning based on the properties of **parallel lines**.

The Proof

1. Given & Construction:

- Start with any triangle ABC.
- **Assumption/Construction:** Draw a line PQ that passes through point A and is **parallel** to the base BC.

2. Logical Steps (Using Angle Relationships):

- **Alternate Interior Angles:** Because line PQ is parallel to BC, the angles created by the "Z" shapes are equal:
  - $\angle PAB = \angle ABC$(let's call this angle B)
  - $\angle QAC = \angle ACB$(let's call this angle C)
- **Straight Line Property:** Points P, A, and Q lie on a straight line. The angles on a straight line must sum to 180^0:
  $\angle PAB + \angle BAC + \angle QAC = 180^0$

3. Conclusion:

- Substitute the equal angles from our logical steps into the equation:
  $\angle B + \angle BAC + C = 180^0$
- Therefore, the three interior angles of the triangle $(A, B$ and $C)$ sum to exactly 180^0.

Why this is Deductive

This proof doesn't rely on measuring a hundred different triangles with a protractor (which would be **inductive**). Instead, it relies on two established "rules" of geometry:

1. Alternate interior angles of parallel lines are equal.
2. Angles on a straight line sum to 180^0.

By following these rules to their logical end, the conclusion becomes an absolute certainty for **all** triangles.

Comparison of Reasoning Types

While the image focuses on deduction, it's helpful to see how it differs from the other common form of mathematical thought:

Type	**Process**	**Certainty**
Deductive	Starts with general rules (definitions) to reach a specific, guaranteed conclusion.	**100% Certain** (if the rules are true).
Inductive	Starts with specific patterns (e.g., 2+4=6, 8+10=18 to guess a general rule $n + (n + 2) = 2n + 2$ for all n.	**Probable**, but requires a formal proof to be certain.

1. Logical Reasoning

Logical reasoning is the foundation of mathematics. It provides a systematic way to analyze statements, determine their truth, and draw valid conclusions. By studying logical reasoning, we develop the ability to construct clear arguments, verify results, and build mathematical proofs in a rigorous and reliable way.

1.1 Propositional logic

Propositional Logic is a branch of logic that studies how statements (called propositions) can be combined and analyzed using logical operations.

Definition: A **proposition (statement)** is a sentence that is either **true (T)** or **false (F)**, but not both.

Examples:

- "2 + 3 = 5" True
- "7 is an even number" False

Non-examples (not propositions):

- "Close the door" (command)
- "What is your name?" (question)

Not all sentences are propositions (statements), however Questions, exclamations, commands, or self-contradictory sentences like the following examples can neither be asserted nor be denied.

- Is mathematics logic?
- Hey there!
- Do not panic.
- This sentence is false.

Sometimes it is unclear whether a sentence identifies a proposition. This can be due to factors such as imprecision or poor sentence structure.

Another example is the sentence

"it is a triangle."

Is this true or false? It is impossible to know because, unlike the other words of the sentence, the meaning of the word it is not determined. In this sentence, the word it is acting like a variable as in $x + 2 = 5$. As the value of x is undetermined, the sentence $x+2 = 5$ is neither true nor false. However, if x represents a particular value, we could make a determination. For example, if $x = 3$, the sentence is true, and if $x = 10$, the sentence is false. Likewise, if it refers to a particular object, then it is a triangle would identify a proposition.

Example

NOT a statement:	Statement:
Add 5 to both sides.	Adding 5 to both sides of $x-5=37$ gives $x=42$.
$\mathbb{Z}$	$42 \in \mathbb{Z}$
42	42 is not a number.
What is the solution of $2x=84$?	The solution of $2x=84$ is 42.

Propositions and Their Types

Note:

Truth (**T**) and falsity (**F**) of a statement is called its truth value.

Simple (Atomic) Statements

A statement is called simple (or atomic) if it cannot be broken down into two or more smaller statements.

Examples:

- 2 is an even number.
- A square has all its sides equal.
- 7 is an odd number.

Compound Statements

A compound statement is a statement formed by combining two or more simple statements.

Example:

"7 is both an odd and a prime number" can be broken into two statements

"7 is an odd number" and "7 is a prime number"

Hence, it is a compound statement.

Note:

1. The simple statements that form a compound statement are called **component statements.**
2. A compound proposition is formed by combining two or more propositions.

   There are six main types of compound propositions:

   - Negation (not)
   - Conjunction (and)
   - Disjunction (or)
   - Exclusive or (either or)
   - Conditional (implication)
   - Biconditional (if and only if)

All of the above are known as logical operators. They are also referred to as logical connectives, with the exception of negation.

More about Propositions

- We use letters such as ***p, q, r,*** and ***s,…*** to represent **propositions**.
- The truth value of a proposition is denoted by T (true) or F (false).

Using these symbols, we can construct new propositions from existing ones. These are the compound propositions. The study of forming and analyzing such propositions is known as **propositional logic** (or **propositional calculus**).
In this context, calculus refers to the manipulation or computation of symbols.

Examples:

Let p: 7 is an odd number

q: 7 is a prime number

Then the compound statement "7 is both an odd and a prime number" can be represented as "*P and q*"

Truth Table

A truth table is a table that displays all possible combinations of truth values of given component propositions and shows the corresponding truth value of a compound proposition for each combination.

Logical Operator

A logical operator (or connective) is a rule that takes one or more propositions as input and produces a new proposition, with its truth value determined by a truth table.

Important logical operators

Which are the building blocks of symbolic logic and discrete mathematics.

Here is a breakdown of the logical operators shown:

Operator	Handle (Natural Language)	Notation (Symbol)	
Negation	*not*	$\neg$	$\sim$
Conjunction	*and*	$\wedge$	
Disjunction	*or*	$\vee$	
Exclusive-or	*xor (either or)*	$\oplus$	
Implication	*implies*	$\Rightarrow$	
Biconditional	*if and only if (iff)*	$\Leftrightarrow$	

The Negation Operator, commonly referred to as **"not"** in logic.

What is Negation?

Negation is a logical operation that takes a statement (p) and flips its truth value. If a statement is true, its negation is false, and vice-versa. It is typically denoted by the symbol $\neg$ (or sometimes $\backsim$).

The Truth Table of Negation

The truth table illustrates how the operator works:

p	$\neg p$
T (True)	**F** (False)
F (False)	**T** (True)

Example1:

Consider the statement $\boldsymbol{p}$: **The number 2 is even.**

Since this statement is **True**, its negation ($\neg \boldsymbol{p}$) must be **False**.

You can express the negation($\neg \boldsymbol{p}$) in several equivalent ways:

- $\neg \boldsymbol{p}$**:** It is **not true** that the number 2 is even.
- $\neg \boldsymbol{p}$**:** It is **false** that the number 2 is even.
- $\neg \boldsymbol{p}$**:** The number 2 is **not** even.

Note:

To negate a proposition, you are essentially stating that the original proposition is "not the case."

Examplex2:

Given the proposition

p: *This book is interesting.*

Its negation $\sim p$ can be read as:

- (i.) This book is **not** interesting.
- (ii.) This book is **uninteresting**.
- (iii.) It is **not the case** that this book is interesting.

Note:

The Negation Operator

The symbol $\sim$ is the **negation operator**. It rules as a **logical complement**, meaning it reverses the truth value of a single proposition. If p is true, $\sim p$ is false, and vice-versa.

Now we introduces the **Conjunction Operator (and)**, a fundamental concept in logic used to combine two statements into a single, new statement.

Key Concepts

- **Definition**: The word "and" is used to join two simple statements (p and q) to form a complex statement, denoted as $p \wedge q$
  .
- **Truth Conditions**: The statement $p \wedge q$ is **true** if and only if both p and q are true. If either (or both) is false, the entire conjunction is false.
- **Symbol**: The symbol ∧ is used to represent the word "and."

Truth Table for Conjunction

A **truth table** summarizes how the truth value of the conjunction depends on the truth values of its components:

p	q	$p \wedge q$
T	T	**T**
F	T	**F**
T	F	**F**
F	F	**F**

Examples:

- **True Statement**: "The number 2 is even and the number 3 is odd." (Both parts are true).
- **False Statement**: "$\overbrace{\text{The number 1 is even}}^{F}$ and $\overbrace{\text{the number 3 is odd}}^{T}$." (The first part is false, making the whole statement false).

1. Disjunction (Inclusive OR: $p \vee q$)

In mathematics, "or" is generally inclusive. This means the statement is true if **at least one** of the propositions is true. It only becomes false if both parts are false.

Truth Table:

p	q	$p \vee q$
T	T	**T**
F	T	**T**
T	F	**T**
F	F	**F**

Example:

- " 3 is even (F) or 4 is odd (F)" is **False** because both parts are false.
- " 4 is even (T) or 3 is odd (T)" is **True** because at least one (in this case, both) is true.

Problems:

1. For the statements

p 15 is odd $\qquad q$: 15 is prime

write each of the following statements as English sentences and determine whether they are true or false. Notice that p is true and q is false.

(a) $p \wedge q$. **(b)** $p \vee q$ **(c)** $p \wedge \neg q$ **(d)** $\neg p \vee \neg q$

2. For the statements

p : 15 is odd $\qquad r$: $15 < 17$

write each of the following statements in symbolic form using the operators $\wedge$, $\vee$, and $\neg$.

(a) $15 \geq 17$.

(b) 15 is odd or $15 \geq 17$.

(c) 15 is even or $15 < 17$.

(d) 15 is odd and $15 \geq 17$.

End of D1